

Singularity of the Φ_3^4 Measure Under Smooth Translations

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1 Introduction

Let μ be the Φ_3^4 measure on the space of distributions $\mathcal{D}'(\mathbb{T}^3)$. Let $\psi \in C^\infty(\mathbb{T}^3)$ be a non-zero smooth function. We define the shift translation $T_\psi : \mathcal{D}'(\mathbb{T}^3) \rightarrow \mathcal{D}'(\mathbb{T}^3)$ by $T_\psi(u) = u + \psi$.

Theorem 1. *The measures μ and $(T_\psi)_*\mu$ are mutually singular.*

2 Integration by Parts and Quasi-Invariance

Assume, for the sake of contradiction, that μ is quasi-invariant under T_ψ . This implies that the pushforward measure $(T_\psi)_*\mu$ is absolutely continuous with respect to μ . Consequently, for the one-parameter family of shifts $T_{t\psi}$, there exists a Radon-Nikodym derivative (density) $\rho_t \in L^1(\mu)$:

$$\frac{d(T_{t\psi})_*\mu}{d\mu}(u) = \rho_t(u).$$

Let F be a smooth cylindrical functional. We compute the expectation with respect to the shifted measure:

$$\mathbb{E}_\mu[F(u + t\psi)] = \int F(v) d((T_{t\psi})_*\mu)(v) = \int F(u) \rho_t(u) d\mu(u).$$

Differentiating with respect to t at $t = 0$:

$$\left. \frac{d}{dt} \right|_{t=0} \mathbb{E}_\mu[F(u + t\psi)] = \mathbb{E}_\mu \left[F(u) \cdot \left. \frac{d}{dt} \right|_{t=0} \rho_t(u) \right].$$

We define the **logarithmic derivative** (or score function) as $B_\psi(u) := \left. \frac{d}{dt} \right|_{t=0} \rho_t(u)$. If μ is quasi-invariant, B_ψ must exist as a random variable in $L^2(\mu)$ (or at least L^1).

3 Finite Dimensional Approximation

We analyze B_ψ via finite-dimensional projections. Let P_N be the projection onto the first N Fourier modes, and let μ_N be the cutoff Φ_3^4 measure on $E_N = P_N L^2(\mathbb{T}^3)$:

$$d\mu_N(u) = \frac{1}{Z_N} \exp(-S_N(u)) du,$$

where the action is $S_N(u) = \int_{\mathbb{T}^3} (\frac{1}{2}u(-\Delta + m^2)u + :P_N(\lambda u^4):_N) dx$, including implicit mass counterterms.

The density of the shifted cutoff measure is given by the change in action:

$$\frac{d(T_{t\psi_N})_*\mu_N}{d\mu_N}(u) = \exp(-[S_N(u - t\psi_N) - S_N(u)]).$$

The finite-dimensional logarithmic derivative is:

$$B_{\psi,N}(u) = -\left. \frac{d}{dt} \right|_{t=0} S_N(u - t\psi_N) = \nabla S_N(u) \cdot \psi_N.$$

The dominant term in the derivative of the ϕ^4 potential is the cubic term. Neglecting the linear Gaussian term (which has finite variance), we have:

$$B_{\psi,N}(u) \approx -4\lambda \int_{\mathbb{T}^3} :u_N^3:(x)\psi(x)dx.$$

Let $X_N := \int_{\mathbb{T}^3} :u_N^3:(x)\psi(x)dx$.

4 Divergence of the Variance

We compute the variance of X_N under the reference Gaussian measure ν_N . By Wick's Theorem:

$$\text{Var}_{\nu_N}(X_N) = 3! \iint_{\mathbb{T}^3 \times \mathbb{T}^3} \psi(x)\psi(y)[C_N(x,y)]^3 dx dy.$$

Small scale asymptotics: In 3D, the covariance $C_N(x,y) \sim |x-y|^{-1}$ for $|x-y| \gg \Lambda_N^{-1}$. Thus, the integrand scales as $|x-y|^{-3}$. Computing the integral in spherical coordinates around the singularity:

$$\int_{\Lambda_N^{-1}}^R r^{-3} \cdot 4\pi r^2 dr \propto \int_{\Lambda_N^{-1}}^R \frac{1}{r} dr = \ln(\Lambda_N R) \sim \ln N.$$

Thus, $\lim_{N \rightarrow \infty} \text{Var}_{\nu_N}(X_N) = \infty$.

5 From Finite to Infinite (Rigorous Justification)

To conclude the proof, we relate the divergence of the Gaussian variance to the interacting variance using the **Inverse Density Argument**.

Lemma 1 (Quantitative Lower Bound via Inverse Density). *Let $\rho_N = \frac{d\mu_N}{d\nu_N}$ be the density of the interacting measure with respect to the Gaussian measure. If the negative moments of the density are uniformly bounded, i.e., $\sup_N \mathbb{E}_{\mu_N}[\rho_N^{-k}] < \infty$ for some $k \geq 1$, then the divergence of the Gaussian variance implies the divergence of the interacting variance.*

Proof. We use the Cauchy-Schwarz inequality to relate moments under different measures. Recall that $\mathbb{E}_{\nu_N}[X_N^2] = \int X_N^2 d\nu_N$. We can rewrite this integral in terms of the interacting measure μ_N by introducing the inverse density:

$$\mathbb{E}_{\nu_N}[X_N^2] = \int X_N^2 \frac{d\nu_N}{d\mu_N} d\mu_N = \mathbb{E}_{\mu_N}[X_N^2 \rho_N^{-1}].$$

Apply the Cauchy-Schwarz inequality to the random variables $A = X_N^2$ and $B = \rho_N^{-1}$ under the measure μ_N :

$$\mathbb{E}_{\mu_N}[X_N^2 \rho_N^{-1}] \leq (\mathbb{E}_{\mu_N}[X_N^4])^{1/2} (\mathbb{E}_{\mu_N}[\rho_N^{-2}])^{1/2}.$$

Substituting the original identity back into the left-hand side:

$$\mathbb{E}_{\nu_N}[X_N^2] \leq (\mathbb{E}_{\mu_N}[X_N^4])^{1/2} (\mathbb{E}_{\mu_N}[\rho_N^{-2}])^{1/2}.$$

Squaring both sides and rearranging to isolate the interacting moment:

$$\mathbb{E}_{\mu_N}[X_N^4] \geq \frac{(\mathbb{E}_{\nu_N}[X_N^2])^2}{\mathbb{E}_{\mu_N}[\rho_N^{-2}]}.$$

We know from Section 4 that $\mathbb{E}_{\nu_N}[X_N^2] \sim \ln N$. Standard stability bounds for the Φ_3^4 measure ensure that the vacuum energy is finite, specifically that $\mathbb{E}_{\mu_N}[\rho_N^{-2}]$ is uniformly bounded in N (or grows at most polynomially). Therefore:

$$\mathbb{E}_{\mu_N}[X_N^4] \geq C(\ln N)^2.$$

This implies that the sequence of random variables X_N does not converge in $L^4(\mu)$, and consequently cannot converge in $L^2(\mu)$ (by hypercontractivity or direct moment comparison). Thus, the variance under the interacting measure diverges. $\square$

Conclusion: Suppose $B_\psi \in L^2(\mu)$ exists. Then, by Fatou's lemma applied to the weak convergence of measures:

$$\|B_\psi\|_{L^2(\mu)}^2 \geq \limsup_{N \rightarrow \infty} \|B_{\psi,N}\|_{L^2(\mu_N)}^2.$$

Since we proved that $\|B_{\psi,N}\|_{L^4(\mu_N)}^4 \rightarrow \infty$, it follows that the limiting random variable cannot have finite moments. This contradicts the assumption that $B_\psi \in L^2(\mu)$. Therefore, the required Radon-Nikodym derivative does not exist, and the measures are singular.